

TADP-Sec Supplementary Material

Trustworthy AI Data Preparation with Governance-to-Runtime Security for Federated Learning

Implementation-aligned supplement for TADP-SEC v14.0

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Purpose of This Supplement

This document provides methodological, experimental, and reproducibility details supporting the accompanying TADP-Sec paper. It is aligned with the final TADP-SEC v14.0 implementation and the final Google Colab Free execution used for the manuscript. It covers the TADP governance rubric and admission policy, the governance-to-security handoff, runtime-profile selection, the complete federated-learning and cryptographic configuration, governance robustness analyses, extended utility and operational results, protected-aggregation fidelity, protocol-enforcement tests, T4 availability, and reproducibility artifacts.

The sensitivity and ablation analyses are post-freeze robustness analyses. They do not retune the governance weights, thresholds, or the canonical admission outcome. T1–T4 are alternative, non-cumulative protection profiles; their tier numbers represent policy requirements rather than a standardized monotonic ranking of cryptographic bit security or implementation cost.

S1. Governance Evidence, Policy, and Controlled Archetypes

S1.1 Governance Dimensions and Scoring

TADP evaluates each candidate contribution across six governance dimensions:

- Source Reliability (SR),
- Data Quality (DQ),
- Documentation (DOC),
- Timeliness (TIME),
- Regulatory Alignment (REG), and
- Context of Use (CTX).

The frozen financial-fraud experimental policy uses the weights below.

All dimensions use a common 0–5 rubric: 0 No Evidence, 1 Very Weak, 2 Limited, 3 Adequate, 4 Strong, and 5 Very Strong. Data owners or stewards provide structured evidence and conditions rather than authoritative numerical trust scores. The server validates the submitted evidence and maps validated conditions to the frozen rubric. DQ is handled differently: it is computed from each client’s local partition by the Local Evidence Validator rather than supplied as a documentary score.

S1.2 Hybrid Provenance Score and Critical Safeguards

Let

$$\mathcal{D} = \{SR, DQ, DOC, TIME, REG, CTX\}.$$

Table 1: Frozen TADP governance weights used in the v14.0 experiment.

Dimension	Weight	Critical
Source Reliability (SR)	0.20	Yes
Data Quality (DQ)	0.25	Yes
Documentation (DOC)	0.15	No
Timeliness (TIME)	0.15	No
Regulatory Alignment (REG)	0.15	Yes
Context of Use (CTX)	0.10	Yes

For client i , the Hybrid Provenance Score (HPS) is

$$HPS_i = \sum_{d \in \mathcal{D}} w_d D_{i,d}, \quad (1)$$

where w_d is the frozen weight and $D_{i,d}$ is the server-validated dimension score.

The critical set is

$$\mathcal{C} = \{SR, DQ, REG, CTX\}, \quad \sum_{d \in \mathcal{C}} w_d = 0.70.$$

The normalized critical score and minimum critical score are

$$HPS_{C,i} = \frac{\sum_{d \in \mathcal{C}} w_d D_{i,d}}{\sum_{d \in \mathcal{C}} w_d}, \quad C_i^{\min} = \min_{d \in \mathcal{C}} D_{i,d}. \quad (2)$$

The frozen v14.0 policy parameters are

$$T_R = 3.0, \quad T_A = 4.0, \quad T_C^{\text{review}} = 3.5, \quad C_{\min} = 3.0,$$

with

$$T_C^{\text{review}} = \frac{T_R + T_A}{2}.$$

There is **no separate** T_C^{direct} **threshold** in the v14.0 policy.

S1.3 Frozen Admission Rule

The implemented rule is deterministic:

1. If $HPS_i < T_R$, the contribution is automatically rejected.
2. If $HPS_i \geq T_R$ but any critical dimension is below $C_{\min}$, the contribution is routed to remediation. A high HPS cannot compensate for a failed critical dimension.
3. If $HPS_i \geq T_A$ and all critical dimensions satisfy $C_{\min}$, the contribution is directly accepted.
4. If $T_R \leq HPS_i < T_A$, the contribution enters automated review and is admitted only if $HPS_{C,i} \geq T_C^{\text{review}}$ and all critical dimensions satisfy $C_{\min}$.
5. Remaining review-band cases are routed to remediation.

The automated-review path uses the already validated governance scores; it is not subjective rescoring, ranking, or quota-based admission.

S1.4 Controlled Governance Evidence

The public ULB credit-card-fraud dataset contains transaction-level features and labels but no organization-level governance documentation. The experiment therefore uses fixed, reproducible archetypes for nontechnical governance evidence while deriving DQ from each actual local partition. The archetypes are experimental fixtures rather than claims about real organizations or universal governance maturity.

Under the frozen policy, TADP admitted 12 of 20 contributors (60%): six by direct acceptance and six through automated review. Five were automatically rejected and three were routed to remediation.

S2. TADP-Sec Security Profile and Session Specification

S2.1 Separation Between Admission and Runtime Protection

TADP and TADP-Sec answer sequential questions. TADP determines whether a contribution is sufficiently governed to participate. Only admitted clients proceed to the TADP-Sec runtime stage, where validated confidentiality, integrity, availability (CIA), and Business Impact (BI) requirements determine the minimum runtime protection profile. CIA/BI requirements are therefore not part of HPS.

S2.2 CIA High-Water and Business-Impact Mapping

For admitted client i ,

$$h_i = \max(C_i, I_i, A_i). \quad (3)$$

The impact levels are Low, Moderate, High, and Critical. Low, Moderate, and High follow impact-based security categorization principles; Critical is a TADP-Sec policy extension. The exact CIA–BI matrix is a framework policy design, not a NIST-prescribed tier matrix.

Table 2: CIA high-water and Business-Impact mapping.

CIA high-water \ BI	Low	Moderate	High	Critical
Low	C1	C2	C3	C4
Moderate	C2	C2	C3	C4
High	C3	C3	C4	C4
Critical	C4	C4	C4	C4

The class-to-profile mapping is

$$C1 \rightarrow T1, \quad C2 \rightarrow T2, \quad C3 \rightarrow T3, \quad C4 \rightarrow T4.$$

S2.3 Session-Wide Protection Policy

Let $\mathcal{A}$ be the closed admitted cohort and T_i the minimum profile required by admitted client i . The server selects

$$T_{\text{session}} = \max_{i \in \mathcal{A}} T_i. \quad (4)$$

The decision is made before federated round 1. The cohort is then closed, the profile is fixed for the session, and it is not downgraded or dynamically changed during the session. A new client, a materially changed dataset commitment, or a material policy change requires reassessment and a new session-security decision. The signed session decision is valid for up to 24 hours; expiry terminates the session and requires a new decision.

In the canonical v14.0 cohort, the 12 admitted clients required T2 for 3 clients, T3 for 4 clients, and T4 for 5 clients. The session requirement was therefore T4. T1–T3 are retained as controlled benchmark profiles over the same admitted cohort; they are not dynamically selected alternatives inside the reported T4-governed session.

S2.4 Runtime Protection Profiles

Table 3: Implementation-backed TADP-Sec runtime profiles.

Tier	Profile	Implemented protection
T1	Governed baseline	Authenticated, session-bound FL with signature verification, freshness/anti-replay checks, Proof-of-Flow/Challenge (PoFC) participation control, and tamper-evident audit logging; no encrypted aggregation.
T2	DF+KS	Modified Domingo-Ferrer fixed-point protected transformation with matrix key switching and protected weighted aggregation.
T3	xMK-CKKS	RLWE-based multi-key encrypted aggregation with an aggregate public key and all-participant collaborative decryption.
T4	SAMK	Multi-key BFV aggregation with Paillier-protected helper information, polynomial interpolation, and independent authorized-client aggregate recovery.

PoFC binds an authenticated submission to a server-issued session/round challenge and enforces the permitted participation/submission limit. It is a protocol-level control, not a network-level denial-of-service defense.

S3. Complete Experimental and Cryptographic Configuration

S3.1 Execution Environment

All final v14.0 experiments were executed on the Google Colab Free tier using a Python 3 CPU runtime with an Intel(R) Xeon(R) CPU at 2.20 GHz, 1 physical core (2 logical CPUs), 12.67 GiB of system RAM, and 107.72 GiB of disk capacity. Runtime values are sequential single-host wall-clock measurements. Generated protocol payloads are implementation-generated logical-role byte counts, not wire-level network captures. Absolute energy and carbon values are not reported because they were not directly instrumented.

S3.2 Dataset and Federated Configuration

The canonical experiment uses the full ULB credit-card-fraud dataset [1]: 284,807 transactions, 492 fraud records, 284,315 normal records, and 29 input features. A stratified 20% split gives 56,962 held-out test records. Twenty percent of the remaining records gives a 45,569-record validation set, leaving 182,276 records for client training; the client-training pool contains 315 fraud records.

The training pool is partitioned across 20 non-IID clients using class-conditional Dirichlet sampling with $\alpha_{\text{fraud}} = 0.80$ and $\alpha_{\text{normal}} = 1.20$. A common StandardScaler is fitted only on the union of client-training records in the simulation and then applied to client, validation, and test records. Validation and test data do not estimate scaler parameters.

The model is regularized logistic regression with 29 feature weights plus a bias term (30 coordinates). Each client trains for 2 local epochs per round using batch size 32, learning rate

0.01, and L2 regularization $\lambda = 0.005$. The server uses sample-size-weighted FedAvg:

$$\alpha_i = \frac{n_i}{\sum_{j \in V_t} n_j}, \quad \Delta_t = \sum_{i \in V_t} \alpha_i \Delta_{i,t}, \quad (5)$$

where V_t is the set of valid current-round contributors. Clients removed before aggregation are excluded from the denominator. The classification threshold is selected on the validation set subject to $\text{FPR} \leq 0.005$ and then applied to the held-out test set.

S3.3 Run Design

Table 4: v14.0 evaluation design.

Study component	Seeds $\times$ rounds	Purpose
B0–T3	5×20	Main controlled comparison
T4 SAMK	3×3	Reference/feasibility evaluation
T1/T2/T3 enforcement tests	3×1	Signature, replay/staleness, and PoFC controls
T4 availability	3×1	Two pre-upload dropouts plus delayed post-upload recovery

B0 and B1 use all 20 clients. A1 and T1–T4 use the same fixed 12-client TADP-admitted cohort.

S3.4 T2: Modified DF with Matrix Key Switching

T2 follows the modified Domingo-Ferrer plus key-switching construction described by Hariss et al. [2]. The v14.0 parameters are

$$d = 25, \quad \lambda = 10, \quad m' = 2^{80}, \quad m = (m')^\lambda = 2^{800}, \quad \text{scale} = 10^6.$$

For each fixed client-key to authorized aggregate-key relation, the public key-switching matrix is prepared once before round 1 and reused. In each protected round the client encrypts the local update and its FedAvg scalar; the logical cloud performs the homomorphic multiplication/expansion, applies the cached key-switching matrix, and aggregates the protected results. Final aggregate decryption occurs only in the designated classified/authorized role. The one-time matrix material is excluded from recurring protocol communication; applying KS is part of per-round cryptographic computation.

DF is a legacy symmetric-key homomorphic construction with known security limitations, including known-plaintext concerns. T2 is therefore a proof-of-concept protected-transformation profile and is not claimed to provide the same standardized security level as T3/T4.

S3.5 T3: xMK-CKKS

T3 uses the xMK-CKKS multi-key workflow [3] with

$$N = 2048, \quad \log_2(Q) = 54, \quad \Delta = 2^{40},$$

nominal RLWE noise standard deviation 3.2 and collaborative decryption-share noise standard deviation 8.0. The parameters are configured to target the Microsoft SEAL tc128 parameter bound for $N = 2048$; the custom reference implementation is not independently certified.

Each admitted client owns an individual secret/public key pair. Participant public keys form one aggregate public key. Clients pack and encrypt the weighted model update, the server adds ciphertexts, distributes the aggregate ciphertext component required for collaborative decryption, and each valid participant returns an aggregate-bound partial decryption share. All valid participant shares are required; the implementation is not a configurable t -of- n threshold scheme.

S3.6 T4: SAMK

T4 follows the SAMK workflow [4] using multi-key BFV, Paillier-protected helper information, and polynomial interpolation. The v14.0 reference parameters are

$$N = 2048, \quad \log_2(q) = 54, \quad t = 16,777,213, \quad \text{scale} = 10^6,$$

with two approximately 1536-bit Paillier primes forming an approximately 3072-bit modulus. The BFV/RLWE parameter profile and 3072-bit Paillier modulus target approximately 128-bit classical security categories, but the Python implementation remains a research/reference implementation rather than a certified production cryptosystem.

Client i first forms its weighted update

$$u_i = \alpha_i \Delta_i$$

and encrypts it with its own BFV key. For ciphertext component $c_{i,1}(X)$ and BFV secret key s_i , the client constructs

$$f_i(X) = s_i c_{i,1}(X). \quad (6)$$

The helper polynomial is evaluated at public interpolation points. The first $N - 1$ helper values are Paillier-encrypted for the server, while the final helper value is separately encrypted for each authorized recipient.

The server aggregates the BFV c_0 components and Paillier-protected helper values, decrypting only the first $N - 1$ aggregate helper points. It retains the final aggregate helper point encrypted for each recipient. Each authorized uploader decrypts its own final point, combines it with the $N - 1$ points, interpolates the aggregate helper polynomial, and recovers the combined weighted FedAvg update. The server does not obtain the plaintext aggregate. The single-process experimental harness may mirror one validated recipient recovery to advance the model; this is not server-side plaintext recovery.

A client unavailable before upload contributes no update and is excluded from the survivor FedAvg denominator. SAMK does not reconstruct a missing update. A client that completed its upload and then became temporarily unavailable can later recover the same already-formed aggregate package.

S3.7 Numerical Fidelity Gates

Protected aggregates are compared with an experiment-side plaintext FedAvg reference using maximum absolute error and root-mean-squared error (RMSE). The reference is used only for validation and is not exposed to the protected aggregation-server interface. Both scenario-specific gates must pass; relative L2 error is diagnostic only.

S4. Governance Robustness and Ablation

S4.1 Scope

The v14.0 robustness analysis loads the stored validated governance records, independently replays the frozen baseline admission policy, verifies reproduction of the stored binary decisions,

Table 5: Scenario-specific protected-aggregation fidelity gates.

Scenario	Max-absolute gate	RMSE gate
B1 xMK-CKKS	2×10^{-4}	5×10^{-5}
T2 DF+KS	2×10^{-6}	5×10^{-7}
T3 xMK-CKKS	2×10^{-4}	5×10^{-5}
T4 SAMK	1×10^{-4}	1×10^{-4}

and only then applies post-freeze perturbations. No governance evidence is rescored or revalidated, no federated training is rerun, and no perturbation is used to retune the reported policy.

S4.2 Threshold Sensitivity

Table 6: Admitted clients under threshold perturbations.

Condition	−30%	−20%	−10%	0%	+10%	+20%	+30%
T_R only	12	12	12	12	11	6	6
T_A only	–	12	12	12	11	6	6
T_R, T_A jointly	12	12	12	12	6	6	3

The −30% upper-threshold-only condition is invalid because it would yield $T_A < T_R$.

S4.3 Weight Sensitivity and LODO

Each governance weight is perturbed one at a time by $\pm 10\%$, $\pm 20\%$, and $\pm 30\%$, with the remaining weights proportionally renormalized. Only one tested condition changed the admitted set: reducing the DQ weight by 30% reduced admission from 12 to 11. All other one-at-a-time tested weight changes retained 12 admitted clients.

Leave-one-dimension-out (LODO) ablation removes one dimension and renormalizes the remainder. Removing DQ reduced admission from 12 to 6, produced six admission flips, and yielded mean $|\Delta HPS| = 0.505$. Removing any other single dimension left the binary admitted set unchanged at 12. These effects describe the controlled v14.0 cohort and do not establish a universal ranking of governance dimensions.

S5. Extended Performance and Security Results

S5.1 Predictive Utility

T4 is compared descriptively with matched T1 trajectories at round 3 because it uses a shorter feasibility horizon. The runtime protection mechanisms preserved the predictive characteristics of their corresponding plain-reference trajectories within the evaluated design.

S5.2 Operational Cost

Protocol communication/round is recurring uplink plus downlink between logical roles. B0/A1/T1 include update uploads and global-model distribution. T2 includes encrypted updates and encrypted FedAvg weights in the uplink, followed by the post-KS aggregate ciphertext and global-model distribution. B1/T3 include xMK ciphertext uploads, distribution of the aggregate component needed for collaborative decryption, returned decryption shares, and global-model

Table 7: Final predictive utility in the v14.0 run.

Scenario	Seeds×Rounds	AP	Δ AP ref.	F1	MCC
B0 Plain FL	5×20	0.7280 ± 0.0002	–	0.7197 ± 0.0000	0.7311 ± 0.0000
B1 xMK-CKKS	5×20	0.7280 ± 0.0002	0 vs B0	0.7197 ± 0.0000	0.7311 ± 0.0000
A1 TADP	5×20	0.7267 ± 0.0003	−0.0013 vs B0	0.7186 ± 0.0026	0.7276 ± 0.0022
T1 Plain TADP	5×20	0.7267 ± 0.0003	0 vs A1	0.7186 ± 0.0026	0.7276 ± 0.0022
T2 DF+KS	5×20	0.7267 ± 0.0003	$+4.93 \times 10^{-9}$ vs T1	0.7186 ± 0.0026	0.7276 ± 0.0022
T3 xMK-CKKS	5×20	0.7267 ± 0.0003	-9.62×10^{-10} vs T1	0.7186 ± 0.0026	0.7276 ± 0.0022
T4 SAMK	3×3	0.7218 ± 0.0010	-2.18×10^{-7} vs T1(R3)	0.7219 ± 0.0209	0.7314 ± 0.0190

Table 8: Operational comparison from the final v14.0 run.

Scenario	Plain/client KB	CT/client KB	Comm./round KB	Crypto/round s	Setup/seed ms	Runtime/seed s
B0 Plain FL	0.240	–	9.60	0.00	0.00	19.6
B1 xMK-CKKS	0.240	28.67	1151.68	10.86	3522.58	240.6
A1 TADP	0.240	–	5.76	0.00	0.00	10.8
T1 Plain TADP	0.240	–	5.76	0.00	0.00	22.9
T2 DF+KS	0.240	75.00	1007.88	4.21	514.35	108.4
T3 xMK-CKKS	0.240	28.67	691.01	6.57	1983.26	156.6
T4 SAMK	0.240	1609.98	19697.01	1145.78	133188.82	3574.4

distribution. T4 includes BFV/Paillier-protected uploads and recipient-specific recovery data. One-time T2 key-switching matrices, xMK public-key setup, and SAMK BFV/Paillier keys are generated before round 1 and reused; their setup traffic is excluded from recurring communication.

Crypto time/round measures the complete protected cryptographic path and excludes one-time setup, local training, model evaluation, and network latency. *Runtime/seed* is end-to-end wall-clock time including setup and all federated rounds. The cost ordering is mechanism-dependent rather than monotonic in tier number.

S5.3 Protected-Aggregation Fidelity

Table 9: Final v14.0 fidelity diagnostics.

Scenario	Checks	Max abs.	Mean RMSE	Max RMSE	Result
B1 xMK-CKKS	100/100	4.29×10^{-7}	9.64×10^{-8}	1.39×10^{-7}	Pass
T2 DF+KS	100/100	9.64×10^{-7}	1.30×10^{-7}	2.06×10^{-7}	Pass
T3 xMK-CKKS	100/100	3.32×10^{-7}	5.52×10^{-8}	8.68×10^{-8}	Pass
T4 SAMK	9/9	4.24×10^{-6}	1.04×10^{-6}	1.26×10^{-6}	Pass

T4 additionally passed all 108/108 authorized-recipient decryptions. All protected scenarios satisfied their scenario-specific maximum-absolute-error and RMSE gates. The fidelity envelope is a numerical acceptance policy for the reference implementation, not a formal cryptographic precision proof.

S5.4 Protocol-Enforcement Tests

All injected violations were blocked before aggregation, and no violating update was aggregated. These tests demonstrate the implemented signature/authentication, freshness/anti-replay, and

Table 10: Protocol-enforcement outcomes (per attack test seed).

Profile	Injected violation	Injected	Blocked	Benign blocked	Effective clients
T1	Invalid signature	2	2/2	0	10
T2	Replay/stale update	1	1/1	0	11
T3	PoFC overflow	3	3/3	0	9

PoFC participation controls. They do not establish robustness against semantic poisoning, backdoors, arbitrary Byzantine behavior, or general network-level denial-of-service attacks.

S5.5 T4 Dropout and Delayed Recovery

The dedicated T4 availability experiment uses three seeds and one round per seed. Two clients are unavailable before upload in each run; they contribute no update and are excluded from the survivor FedAvg denominator. The remaining ten clients form the protected aggregate without downgrading the selected T4 profile. A post-upload delayed recipient is then exercised in each run.

Table 11: T4 availability results.

Measure	Result
Run design	3 seeds $\times$ 1 round
Pre-upload dropouts per run	2
Final effective clients	10
Delayed-recipient recovery	3/3
All authorized recipients passed	Yes
Protected aggregate fidelity passed	Yes
Profile downgraded	No
Server obtained plaintext aggregate	No

A client unavailable before upload is not reconstructed. A client that uploaded before temporary disconnection can recover the same already-formed aggregate after reconnecting. The experiment evaluates a property of the reference workflow, not a production service-level availability guarantee.

S6. Metric Equivalence and Reproducibility

S6.1 Full-Precision Metric-Equivalence Diagnostics

Rounded AP, F1, and MCC can hide small numerical differences between protected and plaintext trajectories. The v14.0 implementation therefore stores full-precision diagnostics, including Δ AP, Δ F1, Δ MCC, final-model parameter L2 distance, maximum test-probability difference, and binary prediction disagreements.

All comparisons produced zero binary prediction disagreements at the selected classification threshold. Thus the protected models are not numerically identical to their references, but the observed numerical perturbations did not change final F1/MCC or binary test predictions in the evaluated runs.

Table 12: Full-precision final diagnostic summary.

Comparison	Mean ΔAP	Max $ \Delta F1 $	Max $ \Delta MCC $	Mean model L2	Max $ \Delta p $
B1 vs B0	0	0	0	9.51×10^{-7}	9.30×10^{-7}
T1 vs A1	0	0	0	0	0
T2 vs T1	4.93×10^{-9}	0	0	1.72×10^{-6}	2.03×10^{-6}
T3 vs T1	-9.62×10^{-10}	0	0	6.38×10^{-7}	1.30×10^{-6}
T4 vs T1(R3)	-2.18×10^{-7}	0	0	8.60×10^{-6}	2.80×10^{-5}

S6.2 AP Non-Inferiority Output

The implementation uses a frozen AP non-inferiority margin $\delta = 0.005$ and writes the planned paired contrasts to `noninferiority_ap_planned_contrasts.csv`. The planned five-seed contrasts are B1 vs B0, A1 vs B0, T1 vs A1, T2 vs T1, and T3 vs T1. T4 is excluded from the planned paired non-inferiority table because its three-seed, three-round feasibility horizon is not directly paired with the five-seed, 20-round main study; T4 is instead reported descriptively against matched T1 round-3 trajectories.

S6.3 Audit and Reproducibility Artifacts

The v14.0 implementation records experiment/protocol versions, dataset identity and commitment information, partition configuration, governance-policy identifiers, evidence-validation outcomes, HPS and critical-profile values, admission decisions, CIA/BI classifications, session-tier decisions, fidelity diagnostics, protocol-enforcement outcomes, T4 availability results, metric-equivalence diagnostics, non-inferiority outputs, and tamper-evident ledger records. The final run produced 22 ledger seals.

The audit layer does not store raw client datasets, private cryptographic keys, or plaintext individual client model updates. The canonical dataset SHA-256 is

76274b691b16a6c49d3f159c883398e03ccd6d1ee12d9d8ee38f4b4b98551a89

The frozen governance profile is `financial_fraud_experimental_profile_v2`, with policy version `2.3-phase1-fixed-threshold-critical-floor-review-profile`.

The implementation reports TADP-SEC v14.0 / TADP-SEC/14.0. The v14.0 source records `TADP_Sec_v13_2.py` as its source-code basis. The source-code basis SHA-256 recorded in the implementation is

56d6833fa655c7aebdd259b23d886c3b88fe76119b3f993851dc4edd39654eb4.

For exact reproduction, the complete v14.0 results archive and the exact executed source should be archived with the submission/repository together with the machine-readable CSV/JSON outputs and audit artifacts.

S7. Scope and Interpretation Notes

- **Governance evidence.** The public ULB dataset does not contain real institutional governance or CIA/BI metadata. Nontechnical conditions therefore use reproducible controlled archetypes, while DQ is technically derived from local client partitions. The experiment demonstrates the governance-to-runtime workflow; it does not establish universally optimal governance weights or thresholds.

- **Cryptographic implementation.** T2–T4 are research/reference implementations. Results demonstrate implementation functionality, numerical fidelity, runtime, generated protocol-object volume, protocol-enforcement behavior, and the evaluated T4 recovery workflow. They are not cryptographic certification of a production deployment.
- **Security level.** T2 is a legacy DF-based proof-of-concept and is not assigned the same standardized security claim as T3/T4. The T3/T4 RLWE parameter profiles and 3072-bit Paillier target approximately 128-bit classical security categories, but the custom implementation is not independently certified.
- **Protocol enforcement.** Signature, replay/staleness, and PoFC tests evaluate specific protocol controls only. They do not establish general poisoning, backdoor, Byzantine, or network-DoS robustness.
- **T4 availability.** Pre-upload unavailable clients are excluded; their missing updates are not reconstructed. Delayed recovery applies only to a client that uploaded before temporary disconnection.
- **Communication and runtime.** Payload values are logical protocol-object byte counts rather than network captures. Runtime is measured on the specified single Colab CPU host and should not be generalized as production latency.

References

References

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